ETP formalization benchmark · abstract-first hints vs experiment 07 arms (no-think, vacuous laws excluded)

Equational Theories Project implications rendered per each run's form, formalized back by each model over OpenRouter, and graded syntactically (`checkform.py`). Correct = exact, or an accepted symmetry (sides swapped, or both equations uniformly dualized).

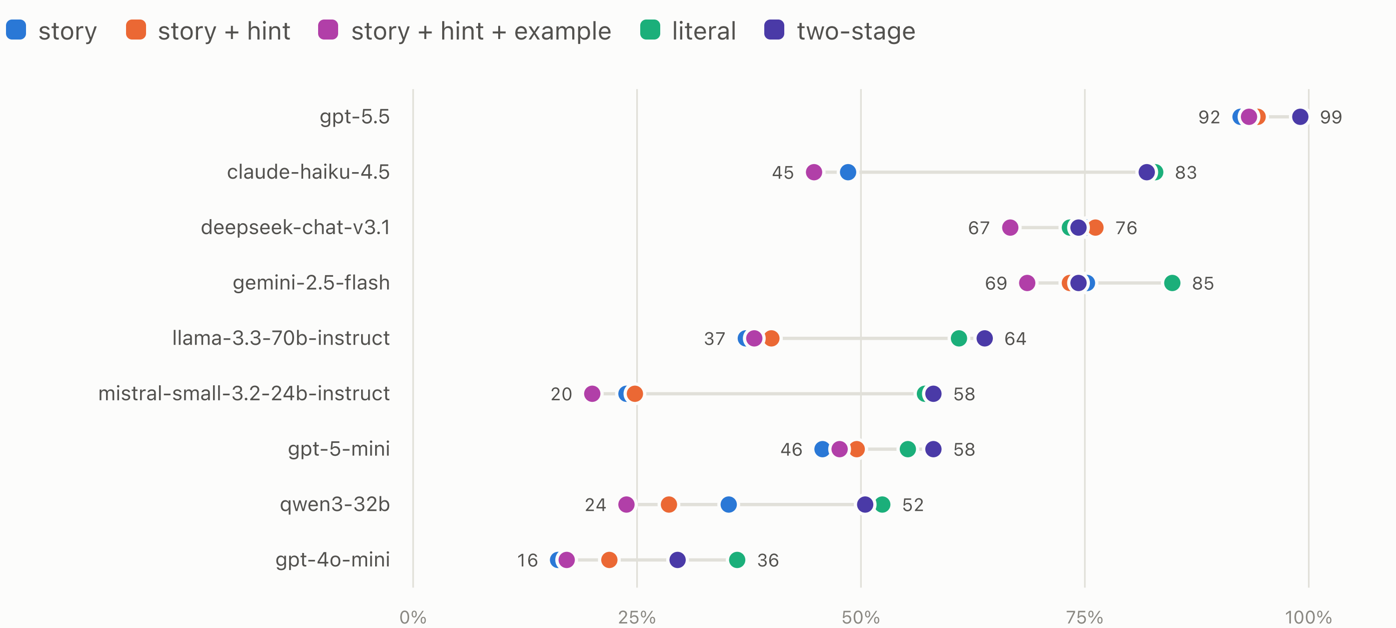
results/no-vacuous/07-two-stage-scale/runs/run-strat20-s0-literal-think-off · results/no-vacuous/07-two-stage-scale/runs/run-strat20-s0-story-think-off · results/no-vacuous/07-two-stage-scale/runs/run-strat20-s0-think-off · experiments/08-abstraction-hint/runs/run-strat20-s0-story-hint-example-think-off · experiments/08-abstraction-hint/runs/run-strat20-s0-story-hint-think-off

Runs	Models	Stories	Graded calls	Overall correct
5	9	105	4,725	56%

105 pairs, 20 per operation-count bin 1–8 minus vacuous-law pairs (bins 2–8 remain) · seed 0 · story vs story + hint vs story + hint + example vs literal vs two-stage · no-think

FIGURE 1 **Correct rate by model and form — no-think · stratified 105**

The same pairs rendered per arm; each row compares one model across the forms under the no-think regime. Sample: 105 pairs — 20 per total-operation bin 1–8, minus the pairs containing a zero-op (vacuous) law; bins 2–8 remain.

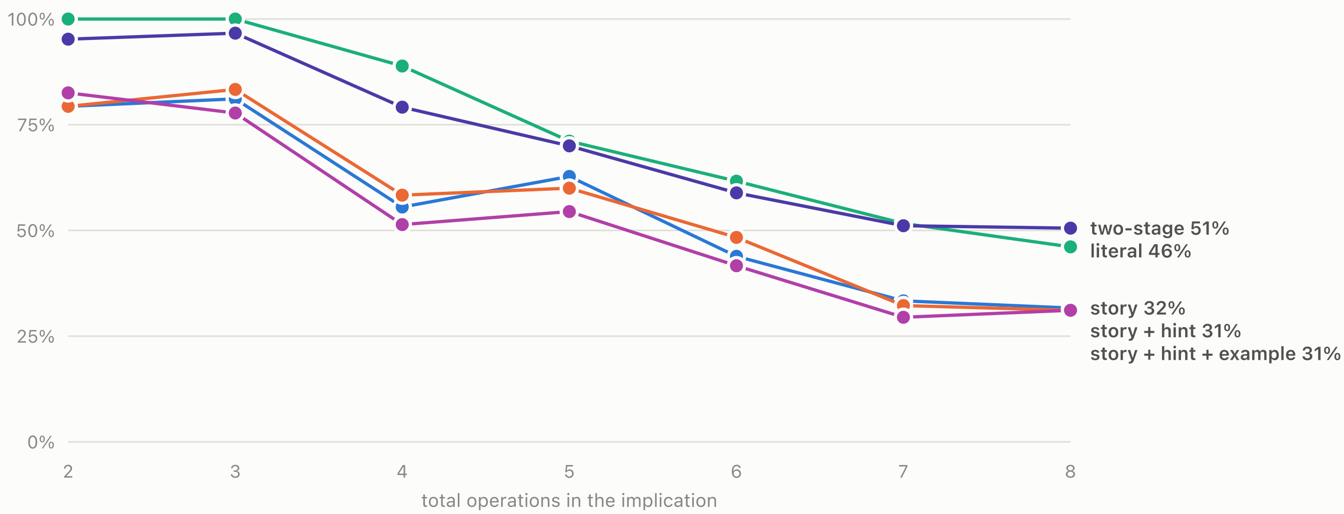


▸ Data: correct rate per model and form

FIGURE 2 **Accuracy vs operation count — no-think · stratified 105**

Correct rate pooled over all models, by the implication's total operation count (both equations combined); one line per form, no-think regime. Sample: 105 pairs — 20 per total-operation bin 1–8, minus the pairs containing a zero-op (vacuous) law; bins 2–8 remain.

story · story + hint · story + hint + example · literal · two-stage

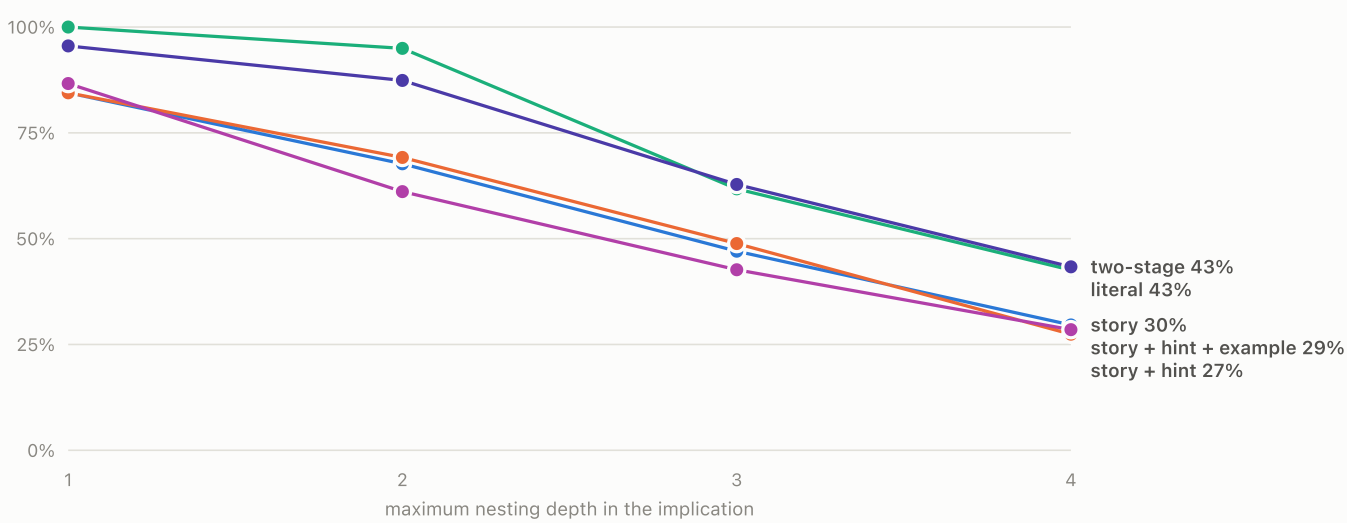


▶ Data: pooled correct rate per operation-count bin

FIGURE 3 **Accuracy vs nesting depth — no-think · stratified 105**

Correct rate pooled over all models, by the deepest operation nesting across the four equation sides; one line per form, no-think regime. Sample: 105 pairs — 20 per total-operation bin 1–8, minus the pairs containing a zero-op (vacuous) law; bins 2–8 remain.

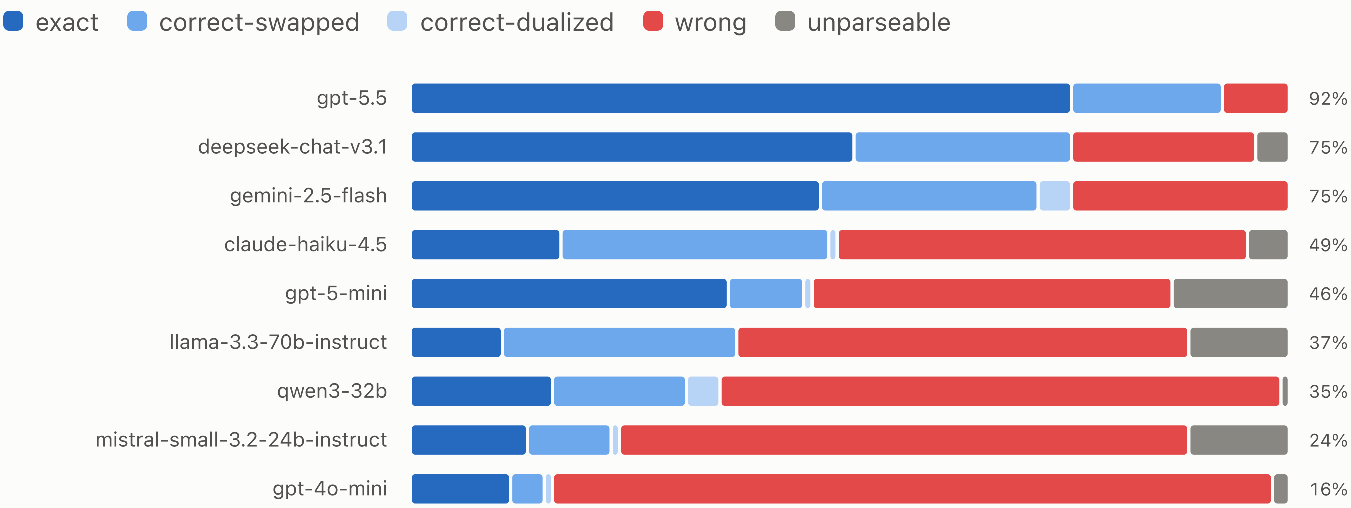
story · story + hint · story + hint + example · literal · two-stage



▶ Data: pooled correct rate per depth bin

FIGURE 4 **Verdict composition — story · no-think · stratified 105**

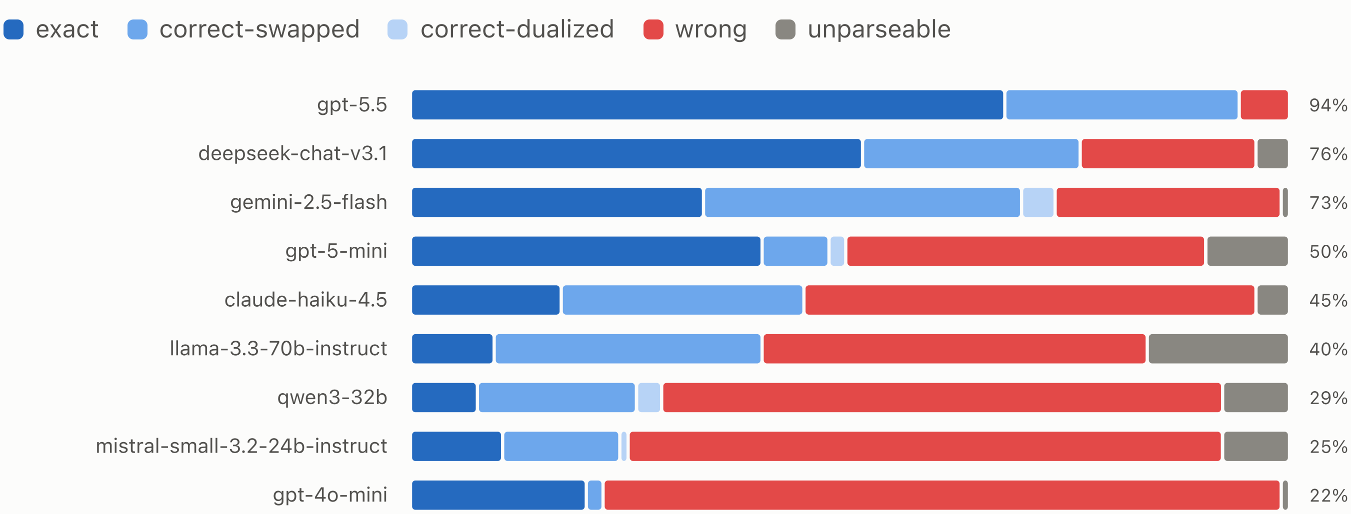
How each model's answers grade out; the right-hand figure is the correct rate (exact + swapped + dualized). Sample: 105 pairs — 20 per total-operation bin 1–8, minus the pairs containing a zero-op (vacuous) law; bins 2–8 remain.



▶ Data: verdict counts per model

FIGURE 5 **Verdict composition — story + hint · no-think · stratified 105**

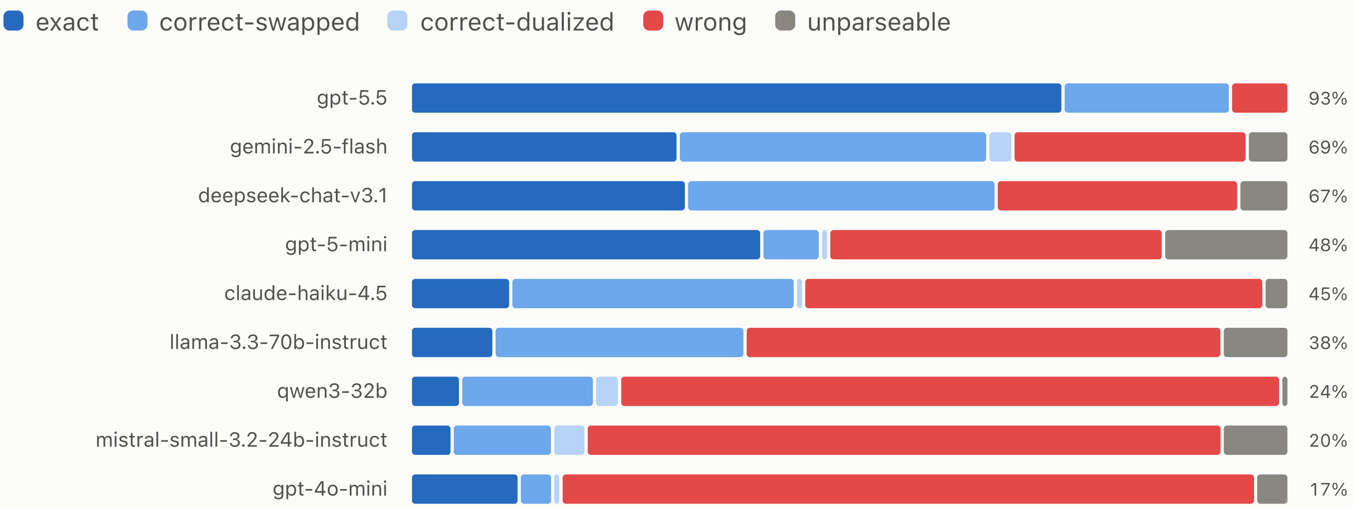
How each model's answers grade out; the right-hand figure is the correct rate (exact + swapped + dualized). Sample: 105 pairs — 20 per total-operation bin 1–8, minus the pairs containing a zero-op (vacuous) law; bins 2–8 remain.



▶ Data: verdict counts per model

FIGURE 6 **Verdict composition — story + hint + example · no-think · stratified 105**

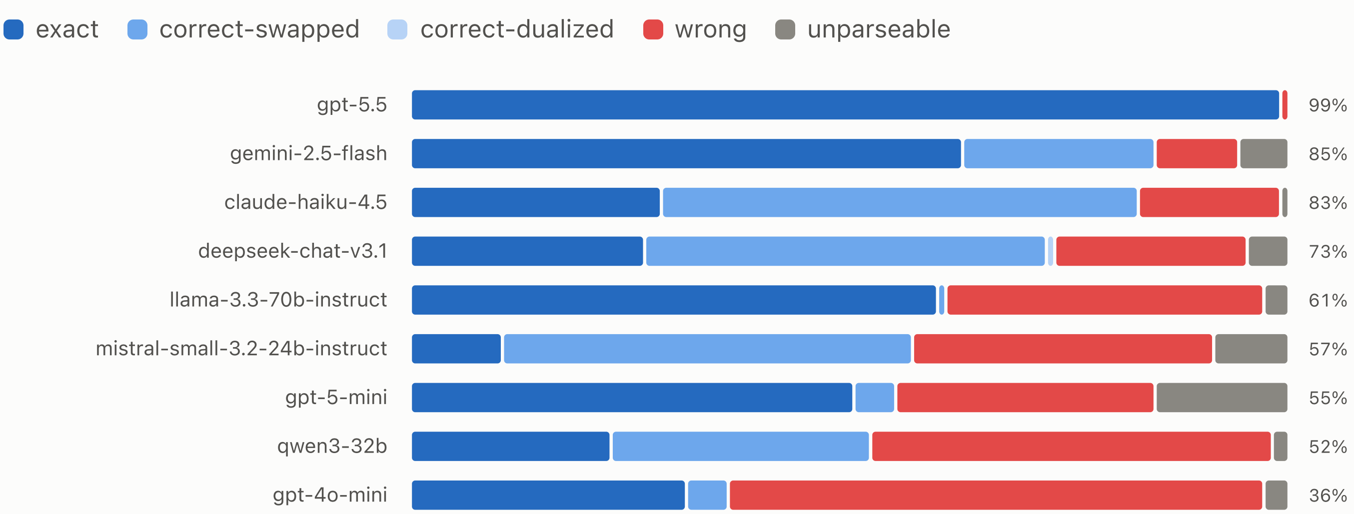
How each model's answers grade out; the right-hand figure is the correct rate (exact + swapped + dualized). Sample: 105 pairs — 20 per total-operation bin 1–8, minus the pairs containing a zero-op (vacuous) law; bins 2–8 remain.



▶ Data: verdict counts per model

FIGURE 7 **Verdict composition — literal · no-think · stratified 105**

How each model's answers grade out; the right-hand figure is the correct rate (exact + swapped + dualized). Sample: 105 pairs — 20 per total-operation bin 1–8, minus the pairs containing a zero-op (vacuous) law; bins 2–8 remain.

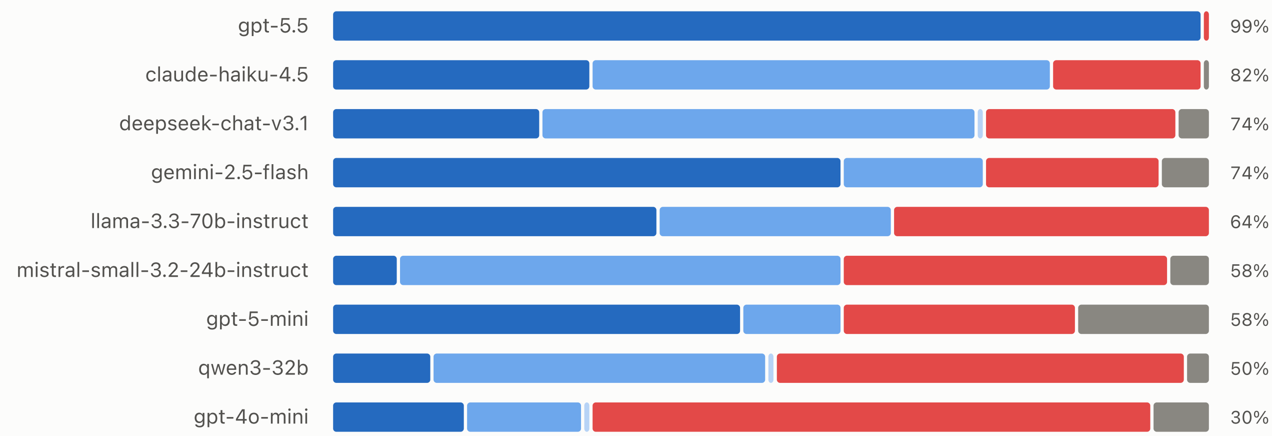


▶ Data: verdict counts per model

FIGURE 8 **Verdict composition — two-stage · no-think · stratified 105**

How each model's answers grade out; the right-hand figure is the correct rate (exact + swapped + dualized). Sample: 105 pairs — 20 per total-operation bin 1–8, minus the pairs containing a zero-op (vacuous) law; bins 2–8 remain.

exact correct-swapped correct-dualized wrong unparseable



▶ Data: verdict counts per model